

Literature Status: Nontrivial Type and Pisier’s Inequality

Run `fa_banach_001`, agent `agent_lane_12`

2026-06-15

Status

This is a lightweight literature-status packet. It records that the main nontrivial-type question in Hytönen–Naor, *Pisier’s inequality revisited*, arXiv:1207.5375, is answered by Ivanisvili–van Handel–Volberg, *Rademacher Type and Enflo Type Coincide*, arXiv:2003.06345.

Source Question

The source paper asks whether every Banach space of nontrivial type satisfies Pisier’s inequality with the logarithmic dimension factor replaced by a constant independent of the cube dimension. In the parsed source this appears at `source.tex:391–395`.

Supporting Answer

The supporting paper gives a finite-cotype characterization. Its Theorem `thm:cotype` states that, for any fixed Banach space X and $1 \leq p < \infty$, Pisier’s inequality holds with a constant independent of dimension if and only if X has finite cotype (`source.tex:411–413` in the parsed supporting source).

Immediately after this theorem, the authors observe that every Banach space with nontrivial type has finite cotype and conclude that Pisier’s inequality therefore holds dimension-free in any Banach space with nontrivial type (`source.tex:414–417`).

Scope

This gives an affirmative later-literature answer to the source question. It does not settle the separate quantitative problem in Hytönen–Naor concerning the correct asymptotic order, as $r \rightarrow \infty$, of the best constant in Pisier’s inequality for $X = \ell_r$.

References

- T. Hytönen and A. Naor, *Pisier’s inequality revisited*, arXiv:1207.5375.
- P. Ivanisvili, R. van Handel, and A. Volberg, *Rademacher Type and Enflo Type Coincide*, arXiv:2003.06345.